

L1a: Course Orientation, Markets, Assets, and the Decision Problem

CHEME 5660 Cornell University

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Objectives for Today

Today we locate the course, argue that you are **already a participant** in the markets it studies, and end on a question we cannot yet answer *rigorously*.

Today's concepts

- **Locate the course:** how four units connect cash-flow models, uncertainty, contingent claims, and algorithmic decisions.
- **Why you are already in this:** most U.S. adults hold a retirement account, and the markets those accounts invest in turn over several hundred times the daily scale of the industries this college trains you to enter.
- **The decision problem:** an instrument is a schedule of dated payments, and two schedules cannot be ranked without a rule for moving dollars across time.

Note: this lecture has no notebook and no notes packet. These slides are the record.

How This Course Runs

Twenty eight lectures in **four units**, each unit adding one piece of machinery to the same underlying question. Six problem sets, spread across the four units, are where the machinery gets used.

What you need

- **Materials**: each week ships as a zip file on the repository releases page. Cloning works too. Released weeks are frozen: fixes are announced, never silently rewritten.
- **Tools**: Julia 1.12 through juliaup, Anaconda for Jupyter, and VS Code with the Julia extension. One setup step, then the notebooks run.
- **Questions**: ask on the course discussion board rather than by email, so an answer reaches everyone who had the same question.

Policies: [grading](#), [attendance](#), [academic integrity](#), and [accommodations](#) are posted in full

What the Semester Feels Like

The policies are posted and you can read them. What a document cannot tell you is the **rhythm**, so here it is.

- **Every lecture builds one thing and uses it immediately.** Miss a week and the next one is harder, not just longer. This is a course with a spine.
- **You will write code.** Not a lot of it, and not from scratch: the course package does the heavy lifting. But a model you have not run is a model you do not believe.
- **The math is the easy part.** Deciding what to model, and admitting when the model is wrong, is the part that takes the semester.

Questions about any of this: now is the time.

You Already Hold a Portfolio

Roughly 61% of U.S. adults hold a retirement account: a 401(k), a 403(b), or an IRA. In household wealth generation, markets are second only to real estate. (2025)

So the question is not whether to participate. You already do.

- That account holds a **portfolio**: a set of positions, in some proportion, chosen by someone.
- Choosing those proportions is an **allocation decision**, and it is the same decision this course spends four units learning to make.
- Most people holding one have **never consciously made it**. A default was selected for them and they left it alone.

Inaction is an allocation. It is just not an argued one.

Twelve People, Eight Meetings a Year

One committee of twelve meets eight times a year and moves the value of everything in that account.

- **Thursday**: you meet them. What they actually control, what they do not, and why that distinction decides which number you use to value anything.
- **September 15 and 16**: they meet. We are three weeks into the course by then, and we come back to it in week four, with the tools to say what the decision did.

That is the whole teaser. Sixty seconds, and we move on.

Link: [FOMC calendar and meeting materials](#)

Why an Engineer

Two industries this college trains people to enter, against the markets next door.
Industry figures are annual, divided by 252 trading days.

	Per year	Cash per trading day
Global biopharma	USD 666B	USD 2.64B
Global batteries	USD 181B	USD 0.72B
Together		USD 3.36B
US debt		USD 1,209B
US equity		USD 608B
Options premium		USD 37B
US markets together		USD 1,854B

A factor of roughly **five hundred**, and the work at that scale is estimation, uncertainty propagation, optimization, and control.

Sources: market sizes 2025 (Fortune Business Insights); turnover 2024–2026 (SIFMA, Cboe)

What That Last Row Controls

One row behaves unlike the others. That **USD 37B** of premium is the cash paid, but it is not the size of the position it opens.

Each contract is an option on **100 shares**. At 60.4 million contracts a day, the shares underneath are worth about **USD 4 trillion**:

$$\frac{\text{USD 37B}}{\text{premium paid}} \bigg/ \frac{\text{USD 4,000B}}{\text{value controlled}} = \frac{0.92\%}{\text{about 92 cents per \$100}}$$

Assume every contract were exercised the day it traded, and that USD 4 trillion of stock changes hands: more than **twice** the debt and equity rows combined. Which is why it is not in the table above.

Careful: $3.9\text{M} \times 100 \times 7,692 = \3.0T of that is one index, settled in cash. It never delivers a share.

Three Kinds of Asset

Everything in this course is one of three things. We name them today and take each apart later, at the point you need it.

Kind	What you own	Where the course takes it apart
Debt	a promise of dated payments	Unit 1, weeks 2 and 3
Equity	a residual claim on a firm	Unit 2, weeks 3 to 7
Derivative	a claim contingent on another asset	Units 3 and 4, weeks 8 onward

Risk and reward scale down the list. Debt promises a schedule and can break it. Equity promises nothing and is worth what is left. A derivative is worth whatever the thing underneath it does, which is the widest range of the three.

Today: we only need the first one, and we will not even use its name.

The Decision Problem

Three kinds of asset, one question. You observe something, you do something, and something comes back. The three words for those are **state**, **action**, and **reward**.

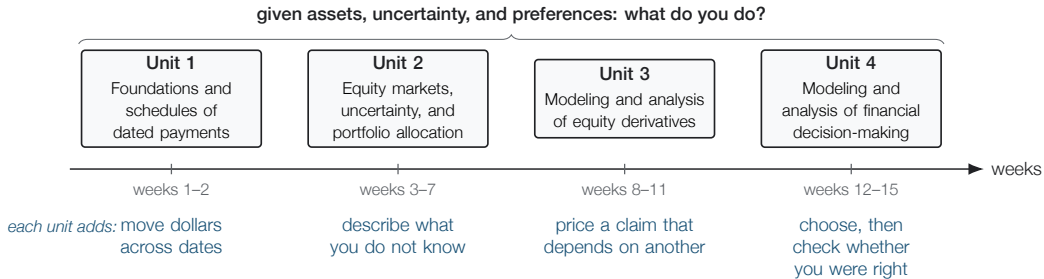


The loop is easy to draw and hard to close. Closing it needs a way to compare payments that arrive on different dates, a way to describe what you do not know, a way to price a claim that depends on something else, and a rule for choosing. Those are the four units.

Careful: the picture is not the problem. Every arrow in it is a modeling choice.

Four Units

Each unit adds exactly one capability, and the next one cannot be built without it.



We start at the left today, and we do not get past it until Thursday.

The Validation Ladder

Four ways to check a claim. Each rung sees data the one below it did not, and each is the **only** rung able to kill its own claim.

	assumptions	past data	current data	money at risk
Real money kills: "I would actually do this"				
Forward test kills: "the market cooperates in real time"				
Backtest kills: "the math works on data that already happened"				
Theory kills: "my reasoning is coherent"				

A **backtest** runs on past data. A **forward test** runs on actual current data with no money at stake, which a brokerage API such as Alpaca makes possible.

Skipping a rung relocates the discovery to where it costs money.

Five Promises

Each costs \$10,000 today, and each is built from a rate the market quoted yesterday. No names yet, just what you are promised.

- A** \$10,028 back, four weeks from today
- B** \$236 every six months for 10 years, then \$10,000
- C** \$264 every six months for **30** years, then \$10,000
- D** \$271 every six months for 10 years, then \$10,000
- E** \$121 every six months for 10 years, plus an inflation adjustment on every payment and on the \$10,000

Which one do you take? Hands up.

As of: published rates for 2026-08-18, each scaled to a \$10,000 outlay

The Names

	What it is called	Annual rate
A	4-week Treasury bill	3.63%
B	10-year Treasury note	4.71%
C	30-year Treasury bond	5.28%
D	investment-grade corporate bonds	5.42%
E	10-year inflation-indexed Treasury (TIPS)	2.41% real

You just reasoned about all five **without knowing any of those words**, because the words are labels for cash-flow schedules. That is all they have ever been, and it is the definition Thursday starts from.

Careful: D is an index of many investment-grade corporate bonds, not one bond. E's rate is *real*, not nominal.

Three Questions We Cannot Answer Yet

Most hands went to **C** or **D**, and that is the right instinct with the information on the previous slide. Here is what it left out.

- **Why not C?** It pays the most of the three government rows. But you may need the money before year thirty, and then you sell at whatever price rates allow. Week 2.
- **Why not D?** It pays the most, full stop. Those borrowers pay **82 basis points** (0.82 percentage points) more than the government for the same timing of money (FRED BAMLC0A0CM, 2026-08-18): the price of not being paid, and of owning something harder to sell. Units 2 and 3.
- **Why not E?** It pays the least of any row, and that comparison is not even legal: its number is in different units. Next slide.

Every row is written as a certain schedule. Only three of them are.

A Forecast by Subtraction

B and E are the same issuer at the same horizon. B promises **nominal** dollars; E promises **real** ones, adjusted for whatever inflation turns out to be. Subtracting one from the other leaves exactly the difference between those two units:

$$\underbrace{4.71\%}_{\text{nominal}} - \underbrace{2.41\%}_{\text{real}} = \underbrace{2.30\%}_{\text{breakeven inflation}}$$

That is the ten-year **breakeven inflation rate**: the inflation rate at which B and E pay you the same. The Federal Reserve publishes it directly, and it prints **2.30%** for the same day. It is not a clean forecast, since it also carries what investors charge for bearing inflation risk, but nothing else gets you this close in one subtraction.

Prices contain forecasts. Extracting them is most of what this course does.

Source: FRED series DGS10, DFII10, T10YIE, as of 2026-08-18

Which One Killed a Bank?

In March 2023 a bank with about \$200 billion in assets failed in two days. It held one of the five promises on your list.

Which one, and why?

None of them defaulted. Every payment that had come due arrived exactly as promised, and the bank failed anyway.

The clip will tell you **which one**. It will not tell you **why** in any way you could check, and that gap is week 2.

Clip: [PBS NewsHour, 28 March 2023, 6 minutes](#)

Thursday

Nobody in this room voted badly. You were not careless and you did not buy something exotic: every promise on that list is issued by the United States government or by large companies that have been paying their bills for decades.

You cannot rank those five yet, and the reason is precise:

You have no rule for comparing dollars that arrive on different dates.

\$264 in thirty years and \$10,028 in four weeks are not comparable numbers until we say what a dollar on one date is worth on another. That rule is Thursday, and it is the first thing this course builds.

Summary

This lecture located the course, argued that you are already a market participant, and read five financial promises as schedules of dated payments.

Key takeaways

- **You are already a portfolio holder.** Retirement accounts make market participation the default, so the allocation question is one nearly everyone answers, most of them by inaction.
- **An instrument is a schedule.** A bill, a note, a bond, a corporate bond, and an inflation-indexed security are labels for schedules of dated payments, though not every schedule is equally certain.
- **A claim is only as good as the rung it survived.** Theory, backtest, forward test, and real money each see different data, and each is the only stage able to falsify its own claim.

On Thursday we build the rule that lets us rank those five schedules.

Further Reading

Not assigned, and not on any problem set. If the sixty seconds on the committee left you curious, start here.

- [*The Structure and Functions of the Federal Reserve System*](#), Federal Reserve Board: the roles of the Board of Governors, the Reserve Banks, and the FOMC.
- [*Economy at a Glance: Policy Rate*](#), Federal Reserve Board: the target range, the effective rate, and the current implementation tools.
- [*Fedpoints: Federal Reserve System*](#), Federal Reserve Bank of New York: an institutional overview.
- [*Role of the Treasury*](#), U.S. Department of the Treasury: fiscal, financial, and debt-management responsibilities. A different institution from the one above, which is week 2's problem.